§13. 多元函数微分学.

1000.13.1

$F(x,y)=\left(\frac{x}{y}\right)^{\frac{1}{x-y}}$ 其中 $x\neq y$. 且 $xy>0$ 设

$$f(x)=\begin{cases}\lim\limits_{y\to x}F(x,y) & x\neq 0\\ e & ;\ x=0\end{cases}$$ $x=0$ 是 $f(x)$ ( ) 间断点、

$$\lim_{y\to x}\left(\frac{x}{y}\right)^{\frac{1}{x-y}}=\lim_{u\to 0}\left(\frac{x}{x+u}\right)^{\frac{1}{-u}}$$

$y-x=u\to 0$

$y=x+u$

$$=\lim_{u\to 0}\left[\left(1-\frac{u}{x+u}\right)^{-\frac{x+u}{u}}\right]^{\frac{1}{x+u}}$$

$$=e^{\frac{1}{y}}\longrightarrow$$ 无穷间断点.

1000.13.2

$f(x,y)=\frac{x}{y^2}e^{-\left(\frac{x}{y}\right)^2},\ y\neq 0,\quad I_1=\lim\limits_{y\to 0}\left[\int_0^1 f(x,y)dx\right]$

$I_2=\int_0^1\left[\lim\limits_{y\to 0}f(x,y)\right]dx.$ 则 ( )

[solution].

$$I_1=\lim_{y\to 0}\left[\int_0^1\frac{x}{y^2}e^{-\left(\frac{x}{y}\right)^2}dx\right]=\lim_{y\to 0}\cancel{\tfrac{1}{2}}\left[e^{-\frac{1}{y^2}}-1\right]^{-\frac{1}{2}}$$

$$\cancel{=\lim_{y\to 0}\frac{1}{4}\cdot\frac{-2}{y^2}}\quad\cancel{\frac{1}{4}}\quad\frac{1}{2}$$

$$I_2=\int_0^1\left[\lim_{y\to 0}\frac{x}{y^2}e^{-\left(\frac{x}{y}\right)^2}\right]dx$$

$$=\int_0^1 0\,dx=0.\qquad\therefore\ I_1>I_2$$

1000.13.3.

$$f(x,y)=\begin{cases}x\sin\frac{1}{y}+y\sin\frac{1}{x}, & xy\neq 0\\ 0 & xy=0.\end{cases}$$

$I_1=\lim\limits_{x\to 0}\left[\lim\limits_{y\to 0}f(x,y)\right],\ I_2=\lim\limits_{\substack{x\to 0\\ y\to 0}}f(x,y).$

[solution]

$$I_1=\lim_{x\to 0}\left[\lim_{y\to 0}\left(x\sin\frac{1}{y}+y\sin\frac{1}{x}\right)\right]$$

$\Downarrow$ ~~当x=0时. = lim~~

$=\cancel{\lim\limits_{x\to 0}}\lim\limits_{y\to 0}x\sin\frac{1}{y}$ 不存在

I_2 存在 $0<\lim\limits_{x,y\to 0}x\sin\frac{1}{y}+y\sin\frac{1}{x}<\lim\limits_{x,y\to 0}|x|+|y|=0$

1000.13.4

$f(x,y)$ 在点 $(0,0)$ 处可微的是

(A). $f(x,y)=\begin{cases}\frac{xy}{x^2+y^2}\\ 0\end{cases}\longrightarrow y=kx$ 不可微

(B). $f(x,y)=\begin{cases}\frac{x^2-y^2}{x^2+y^2}\\ 0\end{cases}\quad y=kx\quad\times$

(C). $f(x,y)=\begin{cases}(x^2+y^2)\sin\frac{1}{\sqrt{x^2+y^2}},\\ 0\end{cases}\quad x\sin\frac{1}{x}\longrightarrow 0$

$$\lim_{(\Delta x,\Delta y)\to(0,0)}f(0+\Delta x,0+\Delta y)-f(0,0)$$

$$=f(\Delta x,\Delta y)=\left[(\Delta x)^2+(\Delta y)^2\right]\sin\frac{1}{\sqrt{(\Delta x)^2+(\Delta y)^2}}$$

$$=0\cdot\Delta x+0\cdot\Delta y+\left[(\Delta x)^2+(\Delta y)^2\right]\sin\frac{1}{\sqrt{\Delta x^2+\Delta y^2}}$$

$$\lim_{(\Delta x,\Delta y)\to(0,0)} \frac{(\Delta x)^2+(\Delta y)^2}{\sqrt{(\Delta x)^2+(\Delta y)^2}} \sin\frac{1}{\sqrt{(\Delta x)^2+(\Delta y)^2}}$$

$$= \lim_{(\Delta x,\Delta y)\to(0,0)} \sqrt{(\Delta x)^2+(\Delta y)^2} \sin\frac{1}{\sqrt{(\Delta x)^2+(\Delta y)^2}} = 0.$$

$\Rightarrow$ (C) 可微

☆ ① $\Delta z = f(x_0+\Delta x_0, y_0+\Delta y_0) - f(x_0, y_0)$

② $A\Delta x + B\Delta y$

③ $\lim_{(\Delta x,\Delta y)\to(0,0)} \frac{\Delta z - (A\Delta x + B\Delta y)}{\sqrt{(\Delta x)^2+(\Delta y)^2}}$

1000. 13. 5.

$f(x,y) = e^{x+y}\left[x^{\frac{1}{3}}(y-1)^{\frac{1}{3}} + y^{\frac{1}{3}}(x-1)^{\frac{2}{3}}\right]$.

则 (0,1) 处的两个偏导数 $f'_x(0,1)$ 和 $f'_y(0,1)$

[Solution].

$$\lim_{\Delta x\to 0} \frac{f(\Delta x,1)-f(0,1)}{\Delta x} = \lim_{\Delta x\to 0} \frac{e^{\Delta x+1}\left[(\Delta x-1)^{\frac{2}{3}}\right] - e[1]}{\Delta x}$$

$$= \lim_{x\to 0} \frac{e^{x+1}\left((x-1)^{\frac{2}{3}}\right)-e}{x} = \lim_{x\to 0} e^{1+x}\left(1+\frac{2}{3}(x-1)^{-\frac{1}{3}}\right) = \frac{e}{3}$$

$$f'_y(0,1) = \left(e^y\left[+y^{\frac{1}{3}}\right]\right)'_y = e^y\left[+y^{\frac{1}{3}} + \frac{1}{3}y^{-\frac{2}{3}}\right]\Big|_{y=1}$$

$$= e\left[+1+\frac{1}{3}\right] = +\frac{4}{3}e$$

1000. 13. 6

$$z = f(x,y) = \begin{cases} \frac{xy^2}{x^2+y^2}, & (x,y)\neq(0,0) \\ 0; & (x,y)=(0,0) \end{cases}$$ 则 $f(x,y)$ 在 (0,0)

(设 $x = ky$ $f(x,y) = \frac{ky^3}{(k+1)y^2}$)

~~$f(x,y) = \frac{x}{1+\frac{x^2}{y^2}}$~~

$\Delta z = f(0+\Delta x, 0+\Delta y) - f(0,0)$

$= f(\Delta x, \Delta y) = \frac{\Delta x(\Delta y)^2}{(\Delta x)^2+(\Delta y)^2}$

$A\Delta x + B\Delta y = 0.$

$\lim_{(\Delta x,\Delta y)\to(0,0)} \frac{\Delta x(\Delta y)^2}{\left[(\Delta x)^2+(\Delta y)^2\right]\sqrt{(\Delta x)^2+(\Delta y)^2}}$ 不存在.

$$\left(\begin{aligned} f(x,y) &= \frac{xy^2}{x^2+y^2} \\ f'_x(x,y) &= \frac{y^2(x^2+y^2) - xy^2\cdot 2x}{(x^2+y^2)^2} = \frac{y^4 - x^2y^2}{(x^2+y^2)^2} \\ f'_y(x,y) &= \frac{2xy(x^2+y^2) - 2xy^3}{(x^2+y^2)^2} = \frac{2x^3y}{(x^2+y^2)^2} \end{aligned} \right)$$

1000. 13.7

设 $f(x,y)$ 连续, $f(0,0)=0$,

$F(x,y) = |x-y|\, f(x,y)$, 则 $F(x,y)$ 在 (0,0) 处.

$$\lim_{x\to 0}\frac{F(x,0)-F(0,0)}{x}=\lim_{x\to 0}\frac{|x|}{x}f(x,0)=0.$$

$\longrightarrow F_x'(0,0)=0.$ 同理 $F_y'(0,0)=0.$

1000.13.8.

$$f(x,y)=\begin{cases}\dfrac{\sqrt{|xy|}}{x^2+y^2}\sin(x^2+y^2)\\ 0\end{cases}\quad (0,0)\text{是?}$$

[solution].

$$f_x'=\lim_{x\to 0}\frac{f(x,0)-f(0,0)}{x-0}=\lim_{x\to 0}\frac{0}{x}=0.$$

$f_y'=0.$

$$\Delta f=f(\Delta x,\Delta y)-f(0,0)=\frac{\sqrt{|\Delta x\Delta y|}}{(\Delta x)^2+(\Delta y)^2}\sin[(\Delta x)^2+(\Delta y)^2]$$

$\cancel{\geq 2\sqrt{|\Delta x\Delta y|}}$

$\cancel{\leq \frac{1}{2}\sin[(\Delta x)^2+(\Delta y)^2]=0.}$

$$\lim_{\rho\to 0(y=x)}\frac{\Delta f-[A\Delta x+B\Delta y]}{\rho}=\frac{\sqrt{|\Delta x\Delta y|}\sin[(\Delta x)^2+(\Delta y)^2]}{(\Delta x^2+\Delta y^2)\sqrt{(\Delta x)^2+(\Delta y)^2}}$$

$0.$

当沿 $y=x$ 趋进时: 原 $=\lim_{\Delta x\to 0}\dfrac{(\Delta x^2)^{\frac12}\sin 2\Delta x^2}{(2\Delta x^2)^{\frac12}\ 2\Delta x^2}=\dfrac{1}{\sqrt2}\neq 0.$

$\therefore f(x,y)$ 在 $(0,0)$ 不可微.

1000.13.9

设 $f(x,y)=|x-y|\varphi(x,y)$, 其中 $\varphi(x,y)$. 其中函 $\varphi(x,y)$ 在 $(0,0)$. 邻域有定义, $f(x,y)$ 在 $(0,0)$ 可微. 充要?

$$f_x'=\begin{cases}\varphi(x,y)+(x-y)\varphi_x'(x,y) & x\geq y.\\ -\varphi(x,y)+(y-x)\varphi_x'(x,y) & x<y\end{cases}$$

f_y' 同理 $\longrightarrow$ 若 $\lim\limits_{(x,y)\to(0,0)}\varphi(x,y)=0\to f_x', f_y'$ 存在.

1000.13.10.

$y=f(x,t)$, 而 t 一 $F(x,y,t)=0$

$\cancel{\dfrac{dy}{dx}=-\dfrac{F_x'}{F_y'}=-}$

$F_t'dt$

全微分. $F_x'dx+F_y'dy+\cancel{F_z'dz}=0.$

① $\to dt=-\dfrac{F_x'}{F_t'}dx-\dfrac{F_y'}{F_t'}dy.$

② $dy=f_x'dx+f_t'dt=f_x'dx-f_t'(\dfrac{F_x'}{F_t'}dx+\dfrac{F_y'}{F_t'}dy).$

$$\Longrightarrow \frac{dy}{dx}=\frac{f_x'F_t'-f_t'F_x'}{f_t'F_y'+F_t'}$$

1000.13.11

$z=f(x,y)=\ln(x+y+\sqrt{1+x^2+y^2})$ 在 $(0,0)$

$f_x(x,y)$ 存在 $f_y'(x,y)$ 存在 $\Longrightarrow$

且 f_x', f_y' 在 $(0,0)$ 连续 $\Rightarrow$ (偏导连续推一切).

1000.13.12

$g(x)$ 可微. $f(x,y)=e^{x-yg(x)}$ $g'(1)=-1$ $\left.\dfrac{\partial f}{\partial x}\right|_{\substack{x=1\\y=1}}=1.$

$\frac{\partial f}{\partial y}\Big|_{\substack{x=1\\y=1}} = ?$

$f(x,y) = e^{x-yg(x)} \longrightarrow$ &

$\frac{\partial f}{\partial x} = e^{x-yg(x)}[1-yg'(x)] \to e^{1-g(1)}[1-(-1)] = 1$

$e^{1-g(1)} = \frac{1}{2}$

$\cancel{\frac{\partial f^2}{\partial x \partial y}} \neq \frac{\partial f}{\partial y} = e^{x-yg(x)}[\cancel{x}-g(x)]$

$e^{1-g(1)}\cdot[1-g(1)] = \frac{1}{2}\cdot(\ln\frac{1}{2}-1)$

$= -\frac{1}{2}(1-\ln\frac{1}{2})$

• 1000.13.13

$u = xe^{-y}z^2$, $z = z(x,y)$ 由方程 $e^{x+y-z}+xyz=1$.

确定，记 $a = \frac{\partial u}{\partial x}\Big|_{(1,0,1)}$. 若该方程也可确定

$y = y(x,z)$. 记 $b = \frac{\partial u}{\partial x}\Big|_{(1,0,1)}$. 则 $a=?$ $b=?$

[solution].

1) $\frac{\partial u}{\partial x} = e^{y}[z^2+2xzz'_x]$

$F(x,y,z) = e^{x+y-z}+xyz-1=0$.

$z_x' = -\frac{F_x'}{F_z'} = +\frac{e^{x+y-z}+yz}{e^{x+y-z}+xy}$

$z'|_{(1,0,1)} = -\frac{e^2}{e^2} = -1$ $\quad \frac{\partial u}{\partial x}\Big|_{(1,0,1)} = 1[1+2(+1)] = \cancel{-1}\,3$

2) $b = \frac{\partial u}{\partial x}$

$= e^{-y}z^2 + xz^2e^{-y}\cdot(-y'_x)\ \cancel{+ xe^{-y}\cdot 2z\, z'_x}$

$y'_x \Rightarrow e^{x+y-z}+xyz=1$

$\xrightarrow{x'} e^{x+y-z}(1+y'_x)+yz+x(zy'_x)=0$.

$\Rightarrow e^{0}(1+y'_x)+0+y'_x=0 \Rightarrow y'_x = -\frac{1}{2}$

$b = 1+1\cdot e^{0}(\frac{1}{2}) + \cancel{1\cdot1\cdot2\cdot1\cdot3} = \frac{3}{2}$.

• 1000.13.14

$u = xy\cdot f(\frac{x+y}{xy})$. $f \neq 0$, 可微, $\quad f \to t \to \begin{matrix} y \\ x\end{matrix}$

$x^2\frac{\partial u}{\partial x} - y^2\frac{\partial u}{\partial y} = G(x,y)u$. $\quad G(x,y) = ?$

$xy - xy - y^2$

$\frac{\partial u}{\partial x} = y\cdot f(\frac{x+y}{xy}) + xy f'_x(\frac{x+y}{xy})\cdot\frac{xy-(x+y)y}{(xy)^2}$

$= yf + \cancel{xy}\, f'_x\cdot\frac{-y^{2}}{xy} = yf - \frac{yf'_x}{x}$ $\quad f'_t$

$\frac{\partial u}{\partial y} = xf - \frac{xf'_y}{y}$ $\quad f'_t$

$x^2\frac{\partial u}{\partial x} - y^2\frac{\partial u}{\partial y} = xy(xf - \frac{yf'_x}{xy} - yf + \frac{xf'_y}{xy})$

$= xy\, G(x,y)$

$G(x,y) = \cancel{x(f-f'_x)-y(f-f'_y)}$ $\quad x-y$

1000.13.15

$z=\frac{x\cos y-y\cos x}{1+\sin x+\sin y}$ $\quad dz\big|_{(0,0)}=?$

$\frac{\partial z}{\partial x}=\frac{(\cos y+y\sin x)(1+\sin x+\sin y)-(x\cos y-y\cos x)(\cos x)}{(1+\sin x+\sin y)^2}\Big|_{(0,0)}=1$

$\frac{\partial z}{\partial y}=\frac{(-x\sin y-\cos x)(1+\sin x+\sin y)-(x\cos y-y\cos x)(\cos y)}{(1+\sin x+\sin y)^2}\Big|_{(0,0)}=-1$

$dz=dx-dy.$

1000.13.16.

$z=x^3+y^3-3x^2-3y^2$ 极小值点?

$\frac{\partial z}{\partial x}=3x^2-6x\overset{令}{=}0$ $\quad x_1=0 \quad x_2=2$ $\quad \frac{\partial^2 z}{\partial x\partial y}=\frac{\partial^2 z}{\partial y\partial x}=0.$

$\frac{\partial z}{\partial y}=3y^2-6y\overset{令}{=}0.$ $\quad y_1=0 \quad y_2=2$

$\cancel{\frac{\partial z}{\partial x}\cdot\frac{\partial z}{\partial y}-\frac{\partial z}{\partial x\partial y}\frac{\partial z}{\partial y\partial x}=3x(x-2)\cdot 3y(y-2)=\Delta}$

$\frac{\partial^2 z}{\partial x^2}=6x-6$ $\quad \frac{\partial^2 z}{\partial y^2}=6y-6.$

$\Delta=(6x-6)(6y-6)-0^2=36(x-1)(y-1).$

$\Delta(0,2)<0$ $\quad \Delta(2,0)<0.$

Δ在$(0,0)<0.$ $\quad \frac{\partial^2 z}{\partial x^2}=-6<0\longrightarrow$ 极大

Δ在$(2,2)>0.$ $\quad \frac{\partial^2 z}{\partial x^2}=6>0\longrightarrow$ 极小.

1000.13.17

$z=\frac{x^2}{2}+xy+\frac{y^2}{2}-2x-2y+5=\underbrace{(x-1)^2+(y-1)^2}_{(1,1)是min}-\frac{1}{2}(x-y)^2+3$

$\frac{\partial z}{\partial x}=x+y-2\overset{令}{=}0$ $\quad \frac{\partial^2 z}{\partial x^2}=1$ $\quad \frac{\partial^2 z}{\partial x\partial y}=1$

$\frac{\partial z}{\partial y}=x+y-2\overset{令}{=}0.$ $\quad \frac{\partial^2 z}{\partial y^2}=1$ $\quad \frac{\partial^2 z}{\partial y\partial x}=1$

$\Delta=0.$

$\longrightarrow \quad x+y=2 \quad y=-x+2$

不妨令$x=1$, $y=1+\Delta y$. ($y=1$, $x=1+\Delta x$ 同理).

$z=0+\Delta y^2-\frac{1}{2}\Delta y^2+3>3.$

令 $x=1+\Delta x$, $y=1+\Delta y$

$z=\Delta x^2+\Delta y^2-\frac{1}{2}(\Delta x^2+\Delta y^2-2\Delta x\Delta y)+3$

$=\frac{1}{2}(\Delta x+\Delta y)^2+3>3.$

$\Rightarrow (1,1)$是极小

1000.13.18

$\lim\limits_{\substack{x\to0\\y\to0}}\frac{f(x,y)-f(0,0)}{x^3+y^3-3x^2-3y^2}=1$

$\frac{f'_x}{3x^2-6x}$ $\quad \frac{f''_{xx}}{6x-6}=1$ 同理 $\frac{f''_{yy}}{6y-6}=1.$

$f''_{xx}=-6$ $\quad f''_{yy}=-6.$

$\longrightarrow$ 极大.

1000.13.19

$\lim_{(x,y)\to(0,0)} \frac{f(x,y)}{(x^2+y^2)^2} = -2$

$\to f(x,y) = (x^2+y^2)^2$

$\frac{\partial f}{\partial x} = 2(x^2+y^2)\cdot 2x$ $\quad \frac{\partial f}{\partial x\partial y} = 4x\cdot 2y = 8xy = \frac{\partial^2 f}{\partial y\partial x}$

$\frac{\partial f}{\partial y} = 2(x^2+y^2)\cdot 2y$

$\frac{\partial^2 f}{\partial x^2} = 4(x^2+y^2) + 4x\cdot 2x$

$= 4(x^2+y^2) + 8x^2$

$\frac{\partial^2 f}{\partial y^2} = 4(x^2+y^2) + 8y^2$.

$\Delta(0,0) = 0$、失效、

~~$f(x,y) = -2\cdot f' + \alpha$~~ 保号性

$f(x,y) = \underbrace{-2(x^2+y^2)^2}_{<0} + \alpha$ $\Rightarrow$ 极大值.

1000.13.20.

$(\frac{\partial u}{\partial x})^2 + (\frac{\partial u}{\partial y})^2 = 1+u^2$ 内部

A. u_{max} min 都在D上取得 ✓ 不能在D内部

则. max 或 min 则 $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0$, 矛盾.

B. u_{max} min 在 D内部取得. ✗

1000.13.21

f, G 可微, $G'_y(x,y) \neq 0$.

(x_0,y_0). f 约束 G 的 极值点.

$\to F(x,y,\lambda) = \lambda G(x,y) + f(x,y)$.

$F'_x = \lambda G'_x + f'_x = 0$

$F'_y = \lambda G'_y + f'_y = 0 \Rightarrow \lambda = -\frac{f'_y}{G'_y}$

$F'_\lambda = G(x_0,y_0) = 0$.

$-\frac{f'_y G'_x}{G'_y} + f'_x = 0 \Rightarrow$ 若 $f'_x \neq 0$ 则 $f'_y \neq 0$

1000.13.22

$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$ 且 $z(x,3x) = x^2$, $z'_1(x,3x) = x^3$.

则 $z''_{12}(x,3x) = ?$

~~$x^2 = \int x^3 dx + \varphi(y) = \frac{x^4}{4} + \varphi(3x)$~~

~~$z - x$, $3x$, x~~

~~$z = x^2$ $\quad z'_1 \cdot 1 = x^3$ $\quad z'_1 \cdot z$~~

$z(x,3x)=\cancel{8}x^2$

$z_1'+3z_2'=2x$

① $\overset{a}{z_{11}''}+3\overset{b}{z_{12}''}+3\overset{b}{z_{21}'}+9\overset{a}{z_{22}''}=2$

$z_1'=x^3$ ✱ $10a+6b=2$

$z_{11}''+3z_{12}''=3x^2$ $a+3b=3x^2$

$\cancel{9a}=8a=2-6x^2$

✱ $(3x^2-\frac{1}{4}+\frac{3}{4}x^2)\frac{1}{3}=b.$ $a=\frac{1}{4}-\frac{3}{4}x^2$

$b=-\frac{1}{12}+\frac{5}{4}x^2$ #

1000.13.23

$F(x,y,z)=f(\pi y-\sqrt{2}z)-e^{2x-z}$ f 可微.

$aF_x'+bF_y'+cF_z'=0$. 则 (a,b,c) 可以是

$aF_x'=f'(\pi y-\sqrt{2}z)\cdot 0-e^{2x-z}2=\underline{-2e^{2x-z}}\cdot a$

$bF_y'=\underline{\pi f'}\cdot b$

$cF_z'=\underline{-\sqrt{2}f'+e^{2x-z}}\cdot c$

$\Rightarrow$ $\pi b-\sqrt{2}c=0$ 若 $a=1$, 则 $c=2$

$-2a+c=0.$ $b=\frac{1}{\pi}\cdot 2\sqrt{2}$ —D.

1000.13.24

$f(x,y)=\begin{cases}\frac{\sin(x^2y)}{2xy}, & xy\neq 0\\ x, & xy=0\end{cases}$, 则 $f_x'(0,1)$

$f_x'=\lim_{\Delta x\to 0}\frac{f(\Delta x,1)-x}{\Delta x}$

$=\lim_{\Delta x\to 0}\frac{\sin(\Delta x^2)-\cancel{0}}{2\Delta x\cdot\Delta x}=\frac{1}{2}$

1000.13.25

$z=(1+x^2y)^{xy^2}$

$x\frac{\partial z}{\partial x}-2y\frac{\partial z}{\partial y}=?$

[solution].

$\frac{\partial z}{\partial x}=\left[e^{xy^2\ln(1+x^2y)}\right]_x'=(1+x^2y)^{xy^2}\cdot\left(y^2\ln(1+x^2y)+\frac{xy^2}{1+x^2y}\cdot 2xy\right)$

$\frac{\partial z}{\partial y}=(1+x^2y)^{xy^2}\left(2xy\ln(1+x^2y)+\frac{(xy^2)\cdot x^2}{1+x^2y}\right)$ x^3y^2 $2x^2y^2$

原 $=(1+x^2y)^{xy^2}\left[(xy^2\cancel{+}-4xy^2)\ln(1+x^2y)+\frac{2x^3y^3-2x^3y^3}{1+x^2y}\right]$

$=(1+x^2y)^{xy^2}(-3xy^2)\ln(1+x^2y)$

131000. 26

$z = z(x,y)$ 由 ~~$z=e^z$~~ $z+e^z=xy$ 所确定

求. $\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{z=0}$?

[solution]

$F = z+e^z-xy=0. \xrightarrow{z=0} xy=1$

$\frac{\partial z}{\partial x} = -\frac{F_x'}{F_z'} = -\frac{-y}{1+e^z} = \frac{y}{1+e^z}$

$\frac{\partial^2 z}{\partial x\partial y} = \frac{\partial\left(\frac{y}{1+e^z}\right)}{\partial y} = \frac{(1+e^z)-ye^z\cdot\frac{\partial z}{\partial y}}{(1+e^z)^2}$

$\frac{\partial z}{\partial y} = -\frac{F_y'}{F_z'} = -\frac{-x}{1+e^z} = \frac{x}{1+e^z}$

$\to$ 原 $= \frac{2-yx/2}{2^2} = \frac{3}{8}$

1000. 13. 27

$f(x,y) = e^{ay}\cos(\ln x) \quad a\neq 0.$

则 $\frac{\partial^2 f}{\partial x^2}+\frac{1}{a^2x^2}\cdot\frac{\partial^2 f}{\partial y^2}+\frac{\partial f}{\partial x} = ?$

$\frac{\partial f}{\partial x} = \frac{-e^{ay}\sin(\ln x)}{x}$ $\quad \frac{\partial^2 f}{\partial x^2} = \frac{\frac{-e^{ay}\cos(\ln x)}{x}x+e^{ay}\sin(\ln x)}{x^2}$

$\frac{\partial f}{\partial y} = ae^{ay}\cos(\ln x)$ $\quad = \frac{e^{ay}(\sin(\ln x)-\cos(\ln x))}{x^2}$

$\frac{\partial f}{\partial x\partial y} = \frac{-ae^{ay}\sin(\ln x)}{x}$ $\quad \frac{\partial^2 f}{\partial y^2} = a^2e^{ay}\cos(\ln x).$

$\frac{\partial^2 f}{\partial x^2}+\frac{1}{a^2x^2}\frac{\partial^2 f}{\partial y^2}+\frac{\partial f}{\partial x}$

$= \frac{e^{ay}(\sin(\ln x)-\cos(\ln x))}{x^2}+\frac{e^{ay}\cos(\ln x)}{x^2}+\frac{-xe^{ay}\sin(\ln x)}{x^2}$

$= \frac{e^{ay}(\sin(\ln x))\cdot(1-x)}{x^2}$

1000. 13. 28.

$z = x\ln\left[(1+y^2)e^{x^2\sin y}\right] = x\ln(1+y^2)+x\ln e^{x^2\sin y}$

$\left(\frac{\partial z}{\partial y} = \frac{x\left[(2y)e^{x^2\sin y}+(1+y^2)e^{x^2\sin y}\cos y\right]}{1+y^2e^{x^2\sin y}}\right)$

$\to z = x\ln(1+y^2)+x^3\sin y$

$\frac{\partial z}{\partial y} = \frac{x\cdot 2y}{1+y^2}+x^3\cos y$

$\frac{\partial^2 z}{\partial y^2} = \frac{2x(1+y^2)-2xy\cdot 2y}{(1+y^2)^2}-x^3\sin y$

$\frac{\partial^3 z}{\partial y^2\partial x} = \frac{2(1+y^2)-4y^2}{(1+y^2)^2}-3x^2\sin y$

$\frac{\partial^4 z}{\partial y^2\partial x^2} = -6x\sin y$

1000. 13. ~~28~~29

$z=z(x,y)$ 由 $x^2y-z=\varphi(x+y+z)$ 确定.
其中 φ 可导, 且 $\varphi'\neq -1$, 则 $\frac{\partial z}{\partial x}=?$

$F=\varphi(x+y+z)-x^2y+z$

$$\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}=-\frac{\varphi'-2xy}{1+\varphi'}$$

1000. 13. 30.

$xyz+\sqrt{x^2+y^2+z^2}=\sqrt{2} \rightarrow z=z(x,y)$.
在 $(1,0,-1)$ 处的全微分? $dz|_{(1,0,-1)}=?$

$F=xyz+\sqrt{x^2+y^2+z^2}-\sqrt{2}=0$.

$$\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}=-\frac{yz+\frac{1}{2}(x^2+y^2+z^2)^{-\frac{1}{2}}\cdot 2x}{xy+\frac{1}{2}(x^2+y^2+z^2)^{-\frac{1}{2}}\cdot 2z}\Bigg|_{(1,0,-1)}=1$$

$$\frac{\partial z}{\partial y}=\frac{xz+\frac{1}{2}(x^2+y^2+z^2)^{-\frac{1}{2}}\cdot 2y}{-(xy+\frac{1}{2}(x^2+y^2+z^2)^{-\frac{1}{2}}\cdot 2\underbrace{z}_{-1})}=\cancel{\frac{\sqrt{2}}{2}}\ -\sqrt{2}$$

$\Rightarrow dz=dx-\sqrt{2}dy$

1000. 13. 31

$z=z(x,y)$ 是由 $e^{x-2y+3z}-2xe^{-y}\cos z=1$ 所确定

$dz|_{(0,0)}=?$ $e^{3z}=1$

$F(x,y,z)=e^{x-2y+3z}-2xe^{-y}\cos z-1=0$.

$$\frac{\partial z}{\partial x}=-\cancel{\frac{F_z'}{F_x'}}=-\frac{F_x'}{F_z'}=-\frac{e^{x-2y+3z}-2e^{-y}\cos z+2xe^{y}\sin z}{3e^{x-2y+3z}+2xe^{-y}\sin z}$$

$$\frac{\partial z}{\partial x}\Big|_{(0,0)}=\frac{e^{3z}-2\cdot 1\cdot 1+0}{-3e^{3z}}=\frac{1}{3}$$

$$\frac{\partial z}{\partial y}\Big|_{(0,0)}=-\frac{F_y'}{F_z'}=-\frac{-2e^{x-2y+3z}+2xe^{-y}\cos z}{3e^{x-2y+3z}+2xe^{-y}\sin z}\Bigg|_{(0,0)}=\frac{-2}{-3}=\frac{2}{3}$$

$$dz|_{(0,0)}=\frac{1}{3}\frac{\partial z}{\partial x}+\frac{2}{3}\frac{\partial z}{\partial y}$$

1000. 13. 32

~~$z=x^t$~~ $z=x^y$, 若 $x=\sin t$. $y=\tan t$. $\frac{dz}{dt}=?$

$z<^{x}_{y}>t$.

$$\frac{dz}{dt}=yx^{y-1}\cdot\cos t+x^y\ln x\cdot\sec^2 t.$$
$$=(\frac{y}{x}\cos t+\ln x\cdot\sec^2 t)x^y$$
$$=(1+\ln x\cdot\sec^2 t)x^y=(1+\frac{\ln\sin t}{\cos^2 t})\cdot(\sin t)^{\tan y}$$

1000. 13. 33

$$f(x,t)=\int_0^{\frac{x}{2\sqrt{at}}}e^{-u^2}du, \quad t \quad a>0.$$

若 $a\frac{\partial^2 f}{\partial x^2}+b\frac{\partial f}{\partial t}=0.\ a>0.\ b=?$

$\frac{\partial f}{\partial t}=e^{-\frac{x^2}{4at}}\cdot\left[\frac{x}{2}(at)^{-\frac{1}{2}}\right]'_t$

$=e^{-\frac{x^2}{4at}}\cdot\frac{x}{2}a^{\frac{1}{2}}\cdot\frac{-1}{2}t^{-\frac{3}{2}}$

$\frac{\partial f}{\partial x}=e^{-\frac{x^2}{4at}}\cdot\frac{1}{2}(at)^{-\frac{1}{2}}$

$\frac{\partial^2 f}{\partial x^2}=e^{-\frac{x^2}{4at}}\cdot\frac{1}{2}(at)^{-\frac{1}{2}}\cdot\left(-\frac{x}{2at}\right)$

$\cancel{+e^{-\frac{x^2}{4at}}}$ $a^{-\frac{1}{2}}$

原 $=e^{-\frac{x^2}{4at}}\cdot\frac{(at)^{-\frac{3}{2}}}{-4}\cdot x a+e^{-\frac{x^2}{4at}}\cdot\frac{-1}{4}b\,a^{\frac{1}{2}}x\,t^{-\frac{3}{2}}$

$\Rightarrow b=-1$

1000.13.34

$z=\frac{y}{f(x^2-y^2)}$, $xy\neq 0$. $\frac{1}{x}\frac{\partial z}{\partial x}+\frac{1}{y}\frac{\partial z}{\partial y}=?$

$\cancel{\frac{\partial z}{\partial x}=}$ $z f(x^2-y^2)-y=F=0$

$\frac{\partial z}{\partial x}=-\frac{F_x}{F_z}=-\frac{zf'\cdot 2x}{f+0}=-\frac{2xzf'}{f}$

$\frac{\partial z}{\partial y}=-\frac{F_y}{F_z}=-\frac{zf'(-2y)-1}{f+0}=-\frac{-2yzf'-1}{f}$

原 $=-\frac{2zf'}{f}+\frac{2zf'}{f}+\frac{1}{yf}=\frac{1}{yf}$

$=\frac{1}{y\cdot\frac{y}{z}}=\frac{z}{y^2}$

1000.13.35.

$F(u,v)$ 对 u,v 二阶连续偏导数.

$z=F(\frac{y}{x},x^2+y^2)$ 则 $\frac{\partial^2 z}{\partial x\partial y}=?$

$\cancel{F=z}$ $F<^{1}_{2}\ \ x,\ y$

$\frac{\partial z}{\partial x}=F_1'\cdot y(-\frac{1}{x^2})+F_2'\cdot 2x$

$\frac{\partial^2 z}{\partial x\partial y}=-\frac{1}{x^2}F_1'-\frac{y}{x^2}\cdot\left(F_{11}''\cdot\frac{1}{x}+F_{12}''\cdot 2y\right)$

$+2x\left(F_{21}''\cdot\frac{1}{x}+F_{22}''\cdot 2y\right)$

1000.13.36.

$z=xy\ln x$ $\quad d(dz)$.

$\frac{\partial z}{\partial x}=y\ln x+y\qquad\frac{\partial z}{\partial y}=x\ln x$

$dz=(y\ln x+y)dx+(x\ln x)dy$

$d(dz)=\left(\frac{y}{x}dx+(1+\ln x)dy\right)dx$

$+\left((1+\ln x)dx+x\ln x\,dydy\right)dy=\frac{y}{x}dx^2+2(\ln x+1)dxdy$

1000.13.37

$z(u,v)$. 阶续

$\cancel{\frac{\partial^2 z}{\partial x\partial y}}$ $\frac{\partial^2 z}{\partial u\partial v}=0$; $u=x^2-y$, $v=f(xy)$.

满足 $f'+xyf''=0$. $\frac{\partial^2 z}{\partial x\partial y}=?$

z < u, v ; x, y.

$$\frac{\partial z}{\partial x} = z_1'(2x) + yz_2'\cdot f_1'$$

$$\frac{\partial^2 z}{\partial x\partial y} = \cancel{2z_1'} + 2x\cdot(-z_{11}'' + \cancel{z_{12}''}) + xy(z_{22}''\cdot f_2')f_1' + yz_2' f_{12}'' + \cancel{y}z_2' f_1'$$

$$= -2x z_{11}'' + xy z_{22}''(f')^2 + z_2'(f'+xyf'')$$

$$= -2x\frac{\partial^2 z}{\partial u^2} + xy(f')^2\frac{\partial^2 z}{\partial v^2}$$

1000.13.38.

f与g均可微, $z=f[xy, \ln x + g(xy)]$

则 $x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y} = ?$

$$\frac{\partial z}{\partial x} = f_1'\cdot y + f_2'\cdot\left(\frac{1}{x} + g'\cdot y\right)$$

$$\frac{\partial z}{\partial y} = f_1'\cdot x + f_2'\cdot(g'\cdot x)$$

原 $= f_2'\cdot(1+xyg') - f_2'(g'xy) = f_2'$

1000.13.39

$\theta, \varphi \in (0, \frac{\pi}{2})$.

$x=\cos\varphi\cos\theta$, $y=\cos\varphi\sin\theta$. $z=\sin\varphi$.

$\frac{\partial^2 z}{\partial x^2}=?$ $F = x^2+y^2+z^2-1=0$.

$$\frac{\partial z}{\partial x} = -\frac{F_x'}{F_z'} = -\frac{2x}{2z} = -\frac{x}{z}$$

$$\frac{\partial^2 z}{\partial x^2} = \cancel{-\frac{1}{z}} - \frac{z - x\frac{\partial z}{\partial x}}{z^2} = -\frac{z+\frac{x^2}{z}}{z^2} = -\frac{x^2+z^2}{z^3}$$

1000.13.40

$z=f(x,y)$

$F(u,v,w)$可微, $F_u'(0,0,0)=1$, $F_v'(0,0,0)=2$

$F_w'(0,0,0)=3$. $F(2x-y+3z, 4x^2-y^2+z^2, xyz)=0$.

$f(1,2)=0$. $f_x'(1,2)=?$

~~$F(x)=$~~ $\frac{\partial z}{\partial x}\Big|_{(1,2)}$ $(x,y,z)=(1,2,0)$.

$2x-y+3z\big|_{(1,2,0)}=0$

$4x^2-y^2+z^2\big|_{(1,2,0)}=0$.

$xyz\big|_{(1,2,0)}=0$.

$$\frac{\partial z}{\partial x} = \frac{-F_x}{F_z} = -\frac{1\cdot 2+2\cdot 8x_0+3\cdot y_0z_0}{1\cdot 3+2\cdot 2z_0+3\cdot x_0y_0}$$

$$= -\frac{2+16+0}{3+6} = -\frac{18}{9} = -2$$

1000.13.41.

$f(x,y)=e^{2x}(x+y^2+2y)$ 极小值?

[solution].

$\frac{\partial f}{\partial x}=2e^{2x}(x+y^2+2y)+e^{2x}\overset{令}{=}0.$

$e^{2x}(\underbrace{2x+2y^2+4y+1}_{=0})=0.$

$\frac{\partial f}{\partial y}=e^{2x}(2y+2)\overset{令}{=}0. \longrightarrow \boxed{\begin{matrix}2x-1=0\\ x=\frac{1}{2}\\ y=-1\end{matrix}}$

$\frac{\partial^2 f}{\partial x^2}=e^{2x}(\underbrace{4x}_{2}+\underbrace{4y^2}_{4}+\underbrace{8y}_{-8}+\underbrace{2+2}_{4})\Big|_{(\frac{1}{2},-1)}=2e>0.$

$\frac{\partial^2 f}{\partial y^2}=2e^{2x}\Big|_{(\frac{1}{2},-1)}=2e$

$\frac{\partial^2 f}{\partial x\partial y}=A \qquad \frac{\partial^2 f}{\partial y\partial x}=B \qquad \sqrt{AB}-\sqrt{4e^2}>0.$ 0

$=e^{2x}(4y+4)$

$=0$

$f(x,y)_{min}=e\left(\frac{1}{2}+1-2\right)=-\frac{e}{2}$

1000.13.42

$f(x,y)=e^{-x}(ax+b-y^2)$ 若 $f(-1,0)$ 为其极大值

a,b 满足:

step1: $\frac{\partial f}{\partial x}=e^{-x}(-ax-b+y^2+a)\overset{(-1,0)}{=}0.$

$e(+a-b+a)=0 \Rightarrow \underline{2a=b}$

$\frac{\partial f}{\partial y}=e^{-x}(-2y)\overset{(-1,0)}{=}0 \Rightarrow e\cdot 0=0$

$\frac{\partial^2 f}{\partial x\partial y}=\frac{\partial^2 f}{\partial y\partial x}=e^{-x}\cdot 2y\overset{(-1,0)}{=}0.$

$\frac{\partial^2 f}{\partial x^2}=e^{-x}(ax+b-y^2-a-a)$

$\overset{(-1,0)}{=}e(-3a+b).$

$\frac{\partial^2 f}{\partial y^2}=-2e^{-x}\overset{(-1,0)}{=}-2e$

$\frac{\partial^2 f}{\partial y^2}\cdot\frac{\partial^2 f}{\partial x^2}-\frac{\partial^2 f}{\partial x\partial y}\frac{\partial^2 f}{\partial y\partial x}=-2e^2(b-3a)>0.$

$3a-b>0. \quad a>0 \quad b>0$

step $\Delta=0$时 $a=b=0.$ $\geq 0.$

$f(x,y)=e^{-x}(-y^2)=\underbrace{-y^2}_{\leq 0}e^{-x}\leq f(-1,0)=0.$

$\therefore 2a=b, \quad a,b\geq 0.$

1000.13.43

$x^2+2y^2+3z^2=1$ 的切平面与三个坐标平面围成的区域最小体积为?

切点 (x_0,y_0,z_0) $\boxed{\frac{a+b+c}{3}\geq\sqrt[3]{abc}}$

~~$A(x-x_0)+B(y-y_0)+C(z-z_0)=0$~~

$F'_x(x-x_0)$ $\quad F'_y \quad F'_z$

$2x_0(x-x_0)+4y_0(y-y_0)+6z_0(z-z_0)=0.$

$\Rightarrow x_0x+2y_0y+3z_0z=1$

$V=\frac{1}{6}\left|\frac{1}{x_0}\cdot\frac{1}{2y_0}\cdot\frac{1}{3z_0}\right|=\frac{1}{6\sqrt{6}}\sqrt{\frac{1}{x_0^2\cdot 2y_0^2\cdot 3z_0^2}}\geq\frac{1}{6\sqrt{6}}\left(\frac{1}{3}\right)^{-\frac{3}{2}}=\frac{\sqrt{2}}{4}$

1000.13.44

长,宽,高. x,y,z. x,y 均以 $1\,cm/s$ 增加,

z 以 $2\,cm/s$ 速率减少, 当 $x=2\,cm$, $y=z=1\,cm$

对角线 l 变化率为?

$$\frac{dl}{dt}=\frac{d(\sqrt{x^2+y^2+z^2})}{dt}=+x(x^2+y^2+z^2)^{-\frac12}\frac{dx}{dt}$$

$$+y(x^2+y^2+z^2)^{-\frac12}\frac{dy}{dt}+z(x^2+y^2+z^2)^{-\frac12}\frac{dz}{dt}$$

$x=2, y=z=1$ 时

$$\frac{dl}{dt}=+2(4+1+1)^{-\frac12}\,1+1(4+1+1)^{-\frac12}\cdot 1-1(4+1+1)^{-\frac12}\,2$$

$$=6^{-\frac12}(+2+1-2)=+\frac{\sqrt6}{6}\ m/s.$$

1000.13.45

设 $z=y^2\ln(1-x^2)$ 求 $\frac{\partial^n z}{\partial x^n}=y^2(n-1)!(-1)^{n-1}[(x+1)^{-n}+(x-1)^{-n}]$

[solution].

$$z=y^2\ln((1-x)(1+x))$$

$$=y^2(\ln(1-x)+\ln(1+x))$$

$$y^2[(x+1)^{-1}+(x-1)^{-1}]$$

$$\frac{\partial z}{\partial x}=\frac{-y^2}{1-x}+\frac{y^2}{1+x}=y^2[-(1-x)^{-1}+(1+x)^{-1}]$$

$$\frac{\partial^2 z}{\partial x^2}=y^2[(-1)(1+x)^{-2}+(-1)^2(-1)(1-x)^{-2}]$$

$$\frac{\partial^n z}{\partial x^n}=y^2[(-1)^{n-1}(1+x)^{-n}+(-1)^{2n-1}(1-x)^{-n}](n-1)!$$

1000.13.46

设 $z=e^{-x}-f(x-2y)$ 且当 $y=0$ 时, $z=x^2$ 求 $\frac{\partial z}{\partial x}$

[solution].

$y=0$ 时: $f(x)=e^{-x}-z=e^{-x}-x^2$

$$\Downarrow$$

$$f(x-2y)=e^{-(x-2y)}-(x-2y)^2$$

$$\frac{\partial z}{\partial x}=-e^{-x}-\frac{\partial f(x-2y)}{\partial x}$$

$$\frac{\partial f}{\partial x}=-e^{-(x-2y)}+2(x-2y)$$

$$\Rightarrow \frac{\partial z}{\partial x}=-e^{-x}-e^{-(x-2y)}+2(x-2y).$$

1000.13.47

$$u(x,y)=\varphi(x+y)+\varphi(x-y)+\int_{x-y}^{0}\psi(t)\,dt$$

其中 φ 有二阶导, ψ 有一阶导.

求 $\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y^2}$

$$\frac{\partial u}{\partial x}=\varphi'(x+y)+\varphi'(x-y)-\psi(x-y)$$

$$\cdot\ \frac{\partial^2 u}{\partial x^2}=\varphi''(x+y)+\varphi''(x-y)-\psi'(x-y)$$

$$\frac{\partial u}{\partial y}=\varphi'(x+y)-\varphi'(x-y)+\psi(x-y)$$

$$\cdot\ \frac{\partial^2 u}{\partial y^2}=\varphi''(x+y)+\varphi''(x-y)-\psi'(x-y)$$

$$\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y^2}=0.$$

1000.13.48.

$z=z(x,y)$ 由方程 $z-y-x+xe^{z}=0$. 确定. $dz=?$

[solution]

$$F(x,y,z)=z-y-x+xe^{z}\equiv 0.$$

$$\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}=-\frac{-1+e^{z}}{1+xe^{z}}$$

$$\frac{\partial z}{\partial y}=-\frac{F_y'}{F_z'}=\frac{-(-1)}{1+xe^{z}}$$

$$dz=\frac{1-e^{z}}{1+xe^{z}}\frac{\partial z}{\partial x}+\frac{1}{1+xe^{z}}\frac{\partial z}{\partial y}$$

1000.13.49

$f(x,y,z)=e^{x}yz^{2}$, $z=z(x,y)$.

是由 $x+y+z+xyz=0$. 确定. 求 $f_x'(0,1,-1)$.

$$F(x,y,z)=x+y+z+xyz$$

而 $$\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}=-\frac{1+yz}{1+xy}$$

$$\frac{\partial z}{\partial y}=-\frac{F_y'}{F_z'}=-\frac{1+xz}{1+xy}$$

$$f_x'=e^{x}yz^{2}+e^{x}y\,2z\cdot\left(-\frac{1+yz}{1+xy}\right)\Bigg|_{(0,1,-1)}$$

$$=1+(-2)\left(-\frac{1-1}{1}\right)=1$$

1000.13.50

$z=\dfrac{y}{f(x^2-y^2)}$, 且 f 可微.

证明 $\dfrac{1}{x}\cdot\dfrac{\partial z}{\partial x}+\dfrac{1}{y}\cdot\dfrac{\partial z}{\partial y}=\dfrac{z}{y^2}$

[solution].

$$\frac{\partial z}{\partial x}=\frac{-y\,f'(x^2-y^2)\cdot 2x}{f^2(x^2-y^2)}$$

$$\frac{\partial z}{\partial y}=\frac{f(x^2-y^2)+y\,f'(x^2-y^2)(2y)}{f^2(x^2-y^2)}$$

$$\frac{-2yf'}{f^2(x^2-y^2)}+\frac{1}{y}\frac{1}{f}+\frac{2yf'}{f^2(x^2-y^2)}=\frac{z}{y^2}$$

1000.13.51

设 $f(u,v)$ 可微, 若 $z=f[x,f(x,x)]$, 求 $\dfrac{dz}{dx}$.

$$\frac{\partial z}{\partial x}=f_1'\cdot 1+f_2'\cdot[f_1'(x,x)+f_2'(x,x)]$$

1000.13.52

$u=f(x,y,z)$ 连续一阶偏导数, $y=y(x)$ 及 $z=z(x)$

由下式确定. $e^{xy}-xy=2$. $z=\int_0^x \sin t^2\,dt$, $\dfrac{du}{dx}$

$$\frac{du}{dt}=\cancel{\frac{\partial u}{\partial x}}\ f_1'\cdot\frac{\partial x}{\partial x}+f_2'\frac{\partial y}{\partial x}\frac{\partial x}{\partial x}+f_3'\cdot\frac{\partial z}{\partial x}\frac{\partial x}{\partial x}$$

$$F(x,y)=e^{xy}-xy-2=0.$$

$$\frac{\partial y}{\partial x}=-\frac{F_x}{F_y}=-\frac{ye^{xy}-y}{xe^{xy}-x}=-\frac{y}{x}\qquad \frac{\partial z}{\partial x}=\sin x^2$$

1000.13.53

$z=\sin(xy)+\varphi(x,\frac{x}{y})$ φ有二阶连续偏导, 求 $\frac{\partial^2 z}{\partial x\partial y}$.

[solution].

$$\frac{\partial z}{\partial x}=y\cos(xy)+\varphi_1'+\varphi_2'\cdot\frac{1}{y}$$

$$\frac{\partial^2 z}{\partial x\partial y}=\cos(xy)-xy\cos(xy)+\varphi_{12}''\cdot\frac{-x}{y^2}+\varphi_{22}''\frac{1}{y}\frac{-x}{y^2}$$

1000.13.54

f(u)有二阶连续导数. $z=xf(\frac{y}{x})+yf(\frac{x}{y})$ 求 $\frac{\partial^2 z}{\partial x\partial y}$

$$\frac{\partial z}{\partial x}=f(\tfrac{y}{x})+xf'(\tfrac{y}{x})\frac{-y}{x^2}+yf'(\tfrac{x}{y})\frac{1}{y}=f(\tfrac{y}{x})-\frac{y}{x}f'(\tfrac{y}{x})+f'(\tfrac{x}{y})$$

$$\frac{\partial^2 z}{\partial x\partial y}=\cancel{\frac{f'(\frac{y}{x})}{x}}-\cancel{\frac{1}{x}f'(\tfrac{y}{x})}-\frac{y}{x^2}f''(\tfrac{y}{x})+f''(\tfrac{x}{y})\frac{-x}{y^2}$$

1000.13.55

$u=yf(\frac{x}{y})+xg(\frac{y}{x})$, f,g 有二阶连续导数.

求 $x\frac{\partial^2 u}{\partial x^2}+y\frac{\partial^2 u}{\partial x\partial y}$

$$\frac{\partial u}{\partial x}=yf'(\tfrac{x}{y})\cdot\frac{1}{y}+xg'(\tfrac{y}{x})\cdot\frac{-y}{x^2}$$

$$=f'-\frac{y}{x}g' \quad +\frac{y}{x^2}g'$$

$$\frac{\partial^2 u}{\partial x^2}=\frac{f''}{y}-\frac{y}{x}g''\frac{-y}{x^2},\quad x\frac{\partial^2 u}{\partial x^2}=\frac{x}{y}f''+\frac{y^2}{x^2}g'' \quad +\frac{y}{x}g'$$

$$\frac{\partial^2 u}{\partial x\partial y}=f''\cdot\frac{-x}{y^2}-\frac{g'}{x}-\frac{y}{x}g''\frac{1}{x}$$

$$y\frac{\partial^2 u}{\partial x\partial y}=-\frac{x}{y}f''-\frac{y}{x}g'-\frac{y^2}{x^2}g''$$

原 = ~~$\frac{y}{x}g'$~~ 0.

1000.13.56

$z=z(x,y)$, 由 $G(x,y,z)=F(xy,yz)=0$ 确定.

$G_z'\neq 0$. 求 $x\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}$

设 $H(x,y,z)=G(x,y,z)-F(xy,yz)=0$.

$$\frac{\partial z}{\partial x}=-\frac{H_x'}{H_z'}=-\frac{G_x'-(F_1'\cdot y+F_2'\cdot 0)}{G_z'-(F_1'\cdot 0+F_2'\cdot y)}=\frac{F_1'y-G_x'}{G_z'-F_2'y}$$

$$\frac{\partial z}{\partial y}=-\frac{H_y'}{H_z'}=-\frac{G_y'-F_1'x-F_2'z}{G_z'-F_2'y}=\frac{F_1'x+F_2'z-G_y'}{G_z'-F_2'y}$$

$$x\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}=\frac{\cancel{xF_1'y}-xG_x'-yzF_2'-yG_y'}{G_z'-F_2'y}=z$$

$G(x,y,z)=0$

$\Rightarrow G_x'=0 \quad G_y'=0 \quad G_z'=0.$

1000.13.57

$x=x(y,z)$, $y=y(z,x)$, $z=z(x,y)$.

均由 $f(x,y,z)=0$ 确定 证 $x_y'\cdot y_z'\cdot z_x'=-1$

[solution].

由 $x=x(y,z)$

又对 $f(x,y,z)=0$ 左右对 y 求偏导: $f_2'+f_1'x_y'=0\Rightarrow x_y'=-\frac{f_2'}{f_1'}$

同理 $y_z'=-\frac{f_3'}{f_2'}$ $z_x'=-\frac{f_1'}{f_3'}$ 得证

1000.13.58.

$u=\varphi(u)+\int_y^x P(t)dt$, $z=f(u)$ $f(u)$. $\varphi(u)$ 可微，$P(t), \varphi'(u)$ 连续

且 $\varphi'(u)\neq 1$. 求 $P(y)\frac{\partial z}{\partial x}+P(x)\frac{\partial z}{\partial y}$.

[solution].

$F(x,y,u)=u-\varphi(u)-\int_y^x P(t)dt$.

$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}=f'\cdot(-\frac{F'_x}{F'_u})=-f'\cdot\frac{P(x)}{1-\varphi'(u)}$

$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial y}=-f'\frac{F_y'}{F_u'}=-f'\frac{-P(y)}{1-\varphi'(u)}$

$\Rightarrow$ 原 $=0$.

1000.13.59

$f(x,y)=\int_0^{xy}e^{-t^2}dt$. 求 $\frac{x}{y}\frac{\partial^2 f}{\partial x^2}-2\frac{\partial^2 f}{\partial x\partial y}+\frac{y}{x}\frac{\partial^2 f}{\partial y^2}$

[solution]

$\frac{\partial f}{\partial x}=e^{-(xy)^2}\cdot y$ $\qquad \frac{\partial^2 f}{\partial x\partial y}=e^{-(xy)^2}+ye^{-(xy)^2}\cdot(-2x^2y)$

$\frac{\partial^2 f}{\partial x^2}=ye^{-(xy)^2}\cdot(-2y^2x)$ $\qquad -2\frac{\partial^2 f}{\partial x\partial y}=-2e^{-(xy)^2}+4x^2y^2e^{-(xy)^2}$

$-x^2y^2$

$\frac{x}{y}\cdot\frac{\partial^2 f}{\partial x^2}=-2y^2x^2\cdot e^{-(xy)^2}$

$\frac{y}{x}\cdot\frac{\partial^2 f}{\partial y^2}=-2y^2x^2\cdot e^{-(xy)^2}$ $\qquad$ plus $=-2e^{-(xy)^2}$

1000.13.60.

$f(x,y)$ 可微，$f(0,0)=0$. $f_x'(0,0)=a$

$f_y'(0,0)=b$ ，且 $\varphi(t)=f[t,f(t,t^2)]$ 求 $\varphi'(0)$.

[solution].

$\cancel{\varphi'(x)=}\varphi'(t)=f_2'[t,f(t,t^2)]\cdot[f_1'(t,t^2)+2tf_2'(t,t^2)]+f_1'[t,f(t,t^2)]$

$\varphi'(t)\Big|_{t=0}=\cancel{f_2'(0,0)}\ b\cdot(a+0)+a=a+ab$

1000.13.61

设 $u=f(x,y,z)$ 有连续偏导数. $y=y(x)$，及 $z=z(x)$.

由 $e^{xy}-y=0$ 和 $e^z-xz=0$ 所确定. 求 $\frac{du}{dx}$

$\frac{du}{dx}=f_1'+f_2'\cdot\frac{\partial y}{\partial x}+f_3'\cdot\frac{\partial z}{\partial x}$. $=\frac{z}{e^z-x}$

$\frac{\partial y}{\partial x}=-\frac{F_x'}{F_y'}=-\frac{ye^{xy}}{xe^{xy}-1}$ $\qquad \frac{\partial z}{\partial x}=-\frac{G_x'}{G_z'}=-\frac{-z}{e^z-x}$

$\frac{du}{dx}=f_1'+f_2'\frac{ye^{xy}}{1-xe^{xy}}+f_3'\frac{z}{e^z-x}$

1000.13.62

$u=f(x,y,z)$. $\varphi(x^2,e^y,z)=0$. $y=\sin x$.

$\frac{du}{dx}=f_x'+f_y'\cdot\cos x+f_z'\cdot\frac{\partial z}{\partial x}$

$\frac{\partial z}{\partial x}=-\frac{2x\varphi_1'+e^{\sin x}\cos x\,\varphi_2'}{\varphi_3'}$

1000.13.63

$$\begin{cases} z=x^2+y^2 \\ x^2+2y^2+3z^2=20 \end{cases}$$ 求 $\frac{dy}{dx}$, $\frac{dz}{dx}$.

[solution]

$$\begin{cases} \frac{dz}{dx} = 2x + 2y\frac{dy}{dx} \\ \cancel{2x}\ 2x+4y\frac{dy}{dx}+6z\frac{dz}{dx}=0 \end{cases} \Rightarrow \begin{cases} \frac{dy}{dx} \\ \frac{dz}{dx} \end{cases}$$

1000.13.64

$f(x,y)$ 在 $O(0,0)$ 处 邻域连续,

$\lim_{(x,y)\to(0,0)} \frac{f(x,y)-xy}{x^2+y^2}=a$ 常数 $a>\frac{1}{2}$, $f(0,0)$是否为$f(x,y)$极值?

[solution]

$$\frac{f(x,y)-xy}{x^2+y^2}=a+\alpha$$

$a=\frac{1}{2}+b \Rightarrow b>0$.

$\Rightarrow f(x,y)= xy+(\frac{1}{2}+b+\alpha)(x^2+y^2)=\frac{1}{2}(x+y)^2+(b+\alpha)(x^2+y^2)$

①: f 连续性 $\Rightarrow \lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$

②. 当$(x,y)\in \mathring{U}_\delta(0)$ 时, $|\alpha|<\frac{b}{2}$.

故在 $\mathring{U}_\delta(0)$ 内 $f(x,y)>0$. $\therefore f(0,0)$ 是$f(x,y)$极小值.

1000.13.65

已知矩形周长为 $2p$, 将它绕其中一边旋转一周而构成旋转体, 求圆柱体半径和高各为多少时, V_{max}.

高 x 半径 $\cancel{r} = p-x$

~~$V=\int_0^x dz\int_0^{p-x} \pi(p-x)^2 r\,dr$~~

$V = x\pi(p-x)^2 = \pi(p^2x+x^3-2px^2)$

$V'_x = \pi(p^2+3x^2-4px)=0$ $\qquad x_0=\frac{4p-2p}{6}=p/3$

~~$V(p)=\pi(2p^3-2p^3)=0$~~ $\qquad x_1=\frac{4p+2p}{2}=3p$

~~$x\in(0,p)$~~ $\quad V(\frac{p}{3})=(\frac{2}{3}p)^2\pi\cdot(\frac{p}{3})=\frac{4p^3}{27}\pi$.

1000.13.66

$x>0, y>0, z>0$. 求 $f(x,y,z)=xyz^3$ 在约束条件 $x^2+y^2+z^2=5R^2$ ($R>0$ 常数) 下最大值.

$F(x,y,z,\lambda) = xyz^3+\lambda(x^2+y^2+z^2-5R^2)$

$$\begin{cases} F'_x = yz^3+2\lambda x \overset{令}{=} 0. \\ F'_y = xz^3+2\lambda y \overset{令}{=} 0 \\ F'_z = 3xyz^2+2\lambda z \overset{令}{=} 0 \\ F'_\lambda = x^2+y^2+z^2-5R^2=0 \end{cases}$$

① $x=y$

~~$x=y=\lambda=0$~~

~~$\begin{cases} x=y=\lambda=0 \\ z=\sqrt{5}R \end{cases}$~~

~~$\begin{cases} z=0 \\ x=y=\sqrt{\frac{5}{2}}R \end{cases}$~~

$\Rightarrow x=y=R \quad z=\sqrt{3}R$

此时 $\max\{f(x,y,z)\} = f(R,R,\sqrt{3}R) = 3\sqrt{3}R^5$

(2) 由1条件证明: 当 $a>0,\ b>0,\ c>0$ 时

$abc^3 \le 27\left(\frac{a+b+c}{5}\right)^5$

$f(a,b,c) = abc^3$

$\max f = 3\sqrt{3}R^5 = 3\sqrt{3}\left(\frac{x^2+y^2+z^2}{5}\right)^{\frac{5}{2}}$

即 $abc^3 \le 27\left(\frac{a+b+c}{5}\right)^5$

1000.13.67

求 $f(x,y) = x+xy-x^2-y^2$ 在闭区域 $D=\{(x,y)\mid 0\le x\le 1,\ 0\le y\le 2\}$ 上的最大值和最小值.

$f(x,y) = x+xy-x^2-y^2$

$\frac{\partial f}{\partial x} = 1+y-2x \quad \frac{\partial^2 f}{\partial x^2} = -2 < 0$

$\frac{\partial f}{\partial y} = x-2y \quad \frac{\partial^2 f}{\partial y^2} = -2 < 0.$

极大值

$\frac{\partial^2 f}{\partial x\partial y} = \frac{\partial^2 f}{\partial y\partial x} = 1 \qquad \Delta = 4-1 = 3 > 0.$

$-3y+1=0 \qquad -2x = -\frac{4}{3}$

令 $\begin{cases} \frac{\partial f}{\partial x} = 0 \to -2x+y+1=0 \\ \frac{\partial f}{\partial y} = 0 \to 2x-4y=0 \end{cases} \to \begin{cases} y = \frac{1}{3} \\ x = \frac{2}{3} \end{cases}$

$6+2-5$

$\Rightarrow$ 极大值: $f(\frac{2}{3},\frac{1}{3}) = \frac{2}{3}+\frac{2}{9}-\frac{1}{9}-\frac{4}{9} = \frac{3}{9} = \frac{1}{3}$

边界上有 极小值. $\frac{-1}{-2}$ $\frac{1}{2}$

$x=0$: $f(0,y) = -y^2 \quad f(0,y)_{min} = -4.$

$x=1$: $f(1,y) = y-y^2 \quad f(1,y)_{min} = -2 \qquad 0,\ 2-4$

$y=0$: $f(x,0) = x-x^2 \quad f(x,0)_{min} = 0 \qquad 0,\ 0$

$y=2$: $f(x,2) = 3x-x^2-4 \qquad -x\ \ \frac{3}{2} = 1.5$

$= -(x^2-3x+4). \qquad -4,\ -2$

$f(x,2)_{min} = -4$

$\Rightarrow$ 极小值: -4.

1000.13.68.

求 $z = x^2+y^2+2x+y$ 在 $D=\{(x,y)\mid x^2+y^2\le 1\}$ max min.

[solution]. $x_0 = -1 \to$ 舍去. $y_0 = -\frac{1}{2}$

$\frac{\partial z}{\partial x} = 2x+2 \quad \frac{\partial z}{\partial y} = 2y+1$

$\frac{\partial^2 z}{\partial x^2} = 2>0 \quad \frac{\partial^2 z}{\partial y^2} = 2>0 \quad \frac{\partial^2 z}{\partial x\partial y} = \frac{\partial^2 z}{\partial y\partial x} = 0.$

$\Delta = 4-0 = 4 > 0. \Rightarrow$ ~~极小值~~.

$z_{(-\frac{1}{2},-\frac{1}{2})} = \frac{1}{4}+\frac{1}{4}-1-\frac{1}{2} = -1$ min. $\qquad z = 1+\sqrt{5}$

$x^2+y^2=1$ 时. $z = 1+2x+\sqrt{1-x^2}$

$z'_x = 2+\frac{1}{2}(1-x^2)^{-\frac{1}{2}}\cdot(-2x)$

$= 2-x(1-x^2)^{-\frac{1}{2}} \overset{令}{=} 0. \qquad x = \frac{2\sqrt{5}}{5} \quad y = \frac{\sqrt{5}}{5}$

$2\sqrt{1-x^2} = x \quad 1-x^2 = \frac{x^2}{4} \quad 5x^2 = 1$

拉格朗日乘子法 ← 在边界上.

$F(x,y,\lambda) = x^2+y^2+2x+y+\lambda(x^2+y^2-1)$

$$\begin{cases} \frac{\partial F}{\partial x} = 2x+2+2\lambda x = 0 \\ \frac{\partial F}{\partial y} = 2y+1+2\lambda y = 0 \\ \frac{\partial F}{\partial \lambda} = x^2+y^2-1 = 0 \end{cases} \longrightarrow x = 2y$$

$\Rightarrow (2x-4y)(1+\lambda) = 0.$

$\begin{cases} \lambda = -1 \\ \cancel{x=} 舍 \end{cases}$ $\begin{cases} y = \frac{\sqrt{5}}{5} \\ x = \frac{2\sqrt{5}}{5} \\ \lambda = \dots \end{cases}$ $\begin{cases} y = -\frac{\sqrt{5}}{5} \\ x = -\frac{2}{5}\sqrt{5} \\ \lambda = \dots \end{cases}$

~~min~~ max $f = 1+\sqrt{5}$ $f = 1-\sqrt{5}$, min ✓

1000.13.69

求内接于椭球面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2} = 1$ $(a,b,c>0)$

的长方体的最大体积.

~~$V=8abc$~~ $V = 8xyz$.

$F(x,y,z,\lambda) = 8xyz - \lambda\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1\right) = 0$.

$F'_x = 8yz - 2\lambda\frac{x}{a^2} = 0.$ $F'_z = 8xy - 2\lambda\frac{z}{c^2} = 0$

$F'_y = 8xz - 2\lambda\frac{y}{b^2} = 0$ $F'_\lambda = \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1 = 0.$

~~易知 $x=y=z$~~

$\Rightarrow$

令 $\frac{x}{a} = u$, $\frac{y}{b} = v$, $\frac{z}{c} = \mu$.

$\Rightarrow \lambda = 8abc\cdot uv\mu.$

$F(u,v,\mu,\lambda) = 8abc\cdot uv\mu - \lambda(u^2+v^2+\mu^2-1) = 0$

$$\begin{cases} F'_u = 8abc\cdot v\mu - 2\lambda u \overset{①}{=} 0 \\ F'_v = 8abc\cdot u\mu - 2\lambda v \overset{②}{=} 0 \\ F'_\mu = 8abc\cdot uv - 2\lambda\mu \overset{③}{=} 0 \\ F'_\lambda = u^2+v^2+\mu^2 = 1 \end{cases} \Rightarrow u = v = \mu.$$

$8abc\cdot v^2 - 2\lambda v = 0.$ $v \neq 0$

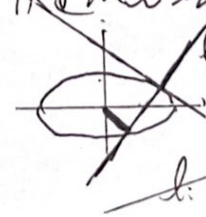
$\rightarrow \begin{cases} 4abc\cdot v = \lambda \\ 3v^2 = 1 \end{cases}$ $\rightarrow \lambda = \frac{4}{3}\sqrt{3}\,abc.$ $\rightarrow v = \frac{\sqrt{3}}{3} = u, \mu.$

$V = 8abc\cdot\frac{3}{27}\sqrt{3} = \frac{8}{9}\sqrt{3}\,abc$ ✓

1000.13.70

第一象限的椭圆 $\frac{x^2}{4}+y^2 = 1$. 求一点,

使过该点法线与原点距离最大.



$\frac{x_0^2}{4}+y_0^2 = 1$ $x_0 = 2\sqrt{1-y_0^2}$

$y = \frac{\sqrt{4-x^2}}{2}$ $y' = -\frac{x_0}{2\sqrt{4-x^2}}$

l: $y - y_0 = \frac{2\sqrt{4-x_0^2}}{x_0}(x-x_0)$

$$y = -\frac{x_0}{2\sqrt{4-x_0^2}}x + y_0 + \frac{x_0^2}{2\sqrt{4-x_0^2}} \quad \frac{2\sqrt{4-x_0^2}}{x_0}x - \frac{3}{2}\sqrt{4-x_0^2}$$

$$y_2 = \frac{2\sqrt{4-x_0^2}}{x_0}x - \frac{x_0}{2\sqrt{4-x_0^2}}x$$

$$\Rightarrow x = \frac{12x_0 - 3x_0^3}{16-3x_0^2} \quad y = \frac{-\frac{3}{2}x_0^2\sqrt{4-x_0^2}}{16-3x_0^2}$$

$$l = x^2+y^2 = \frac{16x_0^2 + 64(4-x_0^2)}{(16-3x_0^2)^2} \quad 12^2x_0^2 - 68x_0^4 + \frac{27}{4}x_0^6$$

$$= \frac{64\times 4 - 48x_0^2}{(16-3x_0^2)^2}$$

$$\overset{令 x_0^2=t}{=} \frac{256-48t}{(16-3t)^2} = \frac{16(16-3t)}{(16-3t)^2} = \frac{16}{16-3t}$$

$$l'_t = 16\cdot(+3)\frac{1}{(16-3t)^2} > 0.$$

$$t_{max} =$$

$$g(x,y) = \frac{x^2}{4} + y^2 - 1.$$

$$\frac{\partial g}{\partial x} = \frac{1}{2}x \quad \frac{\partial g}{\partial y} = 2y.$$

法线: $\frac{X-x}{\frac{1}{2}x} = \frac{Y-y}{2y} \Rightarrow -\frac{1}{2y}Y + \frac{2}{x}X - \frac{3}{2} = 0.$

$$\Rightarrow d = \frac{\frac{3}{2}}{\sqrt{(-\frac{1}{2y})^2 + (\frac{2}{x})^2}}$$

记 $f(x,y) = \frac{1}{4y^2} + \frac{4}{x^2}$

$g(x,y) = \frac{x^2}{4} + y^2 - 1.$

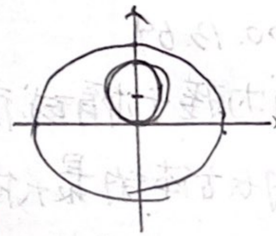
$$F(x,y,\lambda) = \frac{1}{4y^2} + \frac{4}{x^2} + \lambda\left(\frac{x^2}{4} + y^2 - 1\right)$$

$$\begin{cases} F'_x = -8x^{-3} + \frac{\lambda}{2}x \overset{?}{=} 0 \\ F'_y = -\frac{1}{2}y^{-3} + 2\lambda y \overset{?}{=} 0 \\ F'_\lambda = \frac{x^2}{4} + y^2 - 1 \overset{?}{=} 0. \end{cases} \rightarrow y = \frac{1}{4}\sqrt{2}x \quad \lambda = \frac{1}{4y^4}$$

$$\cdots \Rightarrow x = \frac{2}{3}\sqrt{6} \quad y = \frac{\sqrt{3}}{3}$$

1000.13.71

求正数 a,b 的值, s.t. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 包含圆 $x^2+y^2=2y$ 且面积最小、



$$g(x,y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0.$$

$x^2 + (y-1)^2 = 1$ 最小值为1

$$L(x,y,\lambda) = \boxed{x^2 + (y-1)^2} + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1\right)$$

$$L'_x = 0 \quad L'_y = 0 \quad L'_\lambda = 0 \Rightarrow \begin{cases} \lambda = -a^2 \\ y_0 = \frac{b^2}{b^2-a^2} \\ x_0^2 = a^2\left[1 - \frac{b^2}{b^2-a^2}\right] \end{cases}$$

(续 1000.13.71).

$x^2+y^2=2y$ $\quad x^2+(y-1)^2=1$ 化成 $x^2+(y-1)^2$ 最小为1 问题.

$L(x,y,\lambda)=x^2+(y-1)^2+\lambda\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)$

$$\begin{cases} L'_x=2x+\frac{2\lambda}{a^2}x=0 & \lambda_0=-a^2 \\ L'_y=2(y-1)+\frac{2\lambda}{b^2}y=0 & y_0=\frac{b^2}{b^2-a^2} \\ L'_\lambda=\frac{x^2}{a^2}+\frac{y^2}{b^2}-1=0 & x_0^2=a^2\left[1-\frac{b^2}{b^2-a^2}\right] \end{cases}$$

由 $f(x_0,y_0)=1 \Rightarrow a^2b^2-a^4-b^2=0$ ($b^2-a^2=0$ 舍去)

椭圆面积 $=\pi ab \Leftarrow min$

$$\begin{cases} F(x)=a^2b^2-a^4-\cancel{a^2}b^2 \\ min\{ab\} \end{cases}$$

$H(a,b,\eta)=ab+\eta(a^2b^2-a^4-b^2)$

$$\left.\begin{array}{l} H'_a=b+2ab^2\cdot\eta-4a^3\eta=0 \\ H'_b=a+2a^2b\cdot\eta-2b\eta=0 \\ H'_\eta=a^2b^2-a^4-b^2=0 \end{array}\right| \begin{array}{l} \Rightarrow b^2=2a^4 \\ 2a^6-3a^4=0 \end{array}$$

$\Rightarrow a=\frac{\sqrt6}{2}$ $\quad b=\frac{3\sqrt2}{2}$ $\quad\Rightarrow A=\pi ab=\frac{3\sqrt3}{2}\pi$.

例 1000.13.72 $\quad$ 用这个找最值关系

x,y,z 为实数, $e^x+y^2+|z|=3$. 证 $e^xy^2|z|\le1$.

① $z\ge0$ 时.
$$\begin{cases} max\{e^xy^2z\} \\ G(x,y,z)=e^x+y^2+z-3=0. \end{cases}$$

$F(x,y,z,\lambda)=e^xy^2z+\lambda(e^x+y^2+z-3)$ $\quad \lambda=-y^2z$

$$\begin{cases} F'_x=e^xy^2z+\lambda e^x=0. \to y^2z+\lambda=0. \\ F'_y=2e^xyz+2\lambda y=0 \to y^2(e^x-z)=0. \\ F'_z=e^xy^2+\lambda=0 \to y^2(3-y^2-2z)=0. \\ F'_\lambda=e^x+y^2+z-3=0 \to e^x=-y^2-z+3. \end{cases}$$

$yz(3-2y^2-z)=0.$

$$\begin{cases} y=0 \\ \lambda=0. \\ e^x+|z|=3. \end{cases} \qquad \begin{cases} y=1 \\ x=0 \\ z=1 \\ \lambda=-1 \end{cases} \qquad \begin{cases} y=-1 \\ x=0 \\ z=1 \\ \lambda=-1 \end{cases}$$

$e^xy^2|z|\le1$ ✓ $\qquad e^xy^2|z|\le1$ ✓

② $z<0$ 时
$$\begin{cases} max\{-e^xy^2z\} \\ G(x,y,z)=e^x+y^2-z-3=0. \end{cases}$$

$F(x,y,z,\lambda)=-e^xy^2z+\lambda(e^x+y^2-z-3).$

$$\begin{cases} F'_x=-e^xy^2z+\lambda e^x=0 \\ F'_y=-2e^xyz+2\lambda y=0 \\ F'_z=e^xy^2+\lambda=0 \\ F'_\lambda=e^x+y^2-z-3=0 \end{cases} \Rightarrow \lambda=y^2z \quad \begin{cases} y^2(e^x+z)=0 \\ y^2(3-y^2+2z)=0. \\ e^x=-y^2+z+3 \\ -yz(-y^2+z+3)=0. \end{cases}$$

$$\begin{cases} y=0,\ \lambda=0. \\ e^x+|z|=3 \end{cases}$$ 符合

$\cancel{y^2=\frac{3}{2}}$

$\cancel{z=-2}$

$\cancel{y=\pm1.}$

(2) $= 3-e^x-y^2 \geq 0$.

$$\begin{cases} \min \{3-e^x-y^2\} \\ f(x,y) = e^x y^2 (3-e^x-y^2) \end{cases}$$

$F(x,y,\lambda) = \lambda e^x y^2 (3-e^x-y^2) + (3-e^x-y^2)$

$\cancel{F'_x} \Rightarrow \begin{cases} x=0 \\ y=1 \end{cases}$ $\begin{cases} x=0 \\ y=-1 \end{cases}$ $\begin{cases} x \leq \ln 3 \\ y=0. \end{cases}$

$f(0,1)=1$ $f(0,-1)=1$ $f(x,0)=0.$

又. 在边界 $e^x+y^2=3$.

有 $f[x,y(x)] = e^x(3-e^x)(3-e^x-3+e^x) = 0.$

$\therefore f(x,y)\ max = 1.$

1000.13.73

设 $u=u(x,y)$ 满足 $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} + k(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}) = 0.$

确定 a,b. 利用变换 $u(x,y) = v(x,y)e^{ax+by}$

将方程变形. 使新方程不含一阶偏导项.

[solution].

$\frac{\partial^2 u}{\partial x^2}$ $\frac{\partial u}{\partial x} = v_1' e^{ax+by} + v \cdot e^{ax+by} \cdot a$

$\frac{\partial^2 u}{\partial x^2} = v_{11}'' e^{ax+by} + v_1' e^{ax+by} \cdot a + v_1' e^{ax+by} a + v \cdot e^{ax+by} \cdot a^2$

$\frac{\partial^2 u}{\partial y^2} = v_{22}'' e^{ax+by} + v_2' e^{ax+by} \cdot b + v_2' e^{ax+by} \cdot b + v \cdot e^{ax+by} \cdot b^2$

$\frac{\partial u}{\partial y} = v_2' e^{ax+by} + v \cdot e^{ax+by} \cdot b.$

原 $= v_{11}'' e^{\bigcirc} - v_{22}'' e^{\bigcirc} + v \cdot e^{\bigcirc} \cdot a^2 - v \cdot e^{\bigcirc} \cdot b^2$

$+ e^{\bigcirc}(v_1' \cdot 2a - v_2' \cdot 2b). + e^{\bigcirc} v_1' k + e^{\bigcirc} v_2' k + \cdots$

$\Rightarrow \begin{cases} 2a+k=0 \\ -2b+k=0 \\ \cancel{a^2-b^2+ak-bk=0}. \end{cases} \Rightarrow a+b=0.$

$\Rightarrow a=-\frac{k}{2}$ $b=\frac{k}{2}$

$\longrightarrow$ 原变成 $\frac{\partial^2 v}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0.$

1000.13.74

设 A,B,C 常数. $AC-B^2<0.$ $A \neq 0.$

$u(x,y)$ 二阶连续偏导数.

证明. 必存在非奇异线性变换.

$\xi = \lambda_1 x + y$ $\quad \eta = \lambda_2 x + y$ (λ_1, λ_2 为常数).

将方程 $A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} = 0$ 化成 $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0.$

[solution]. $u < \begin{matrix} \xi \\ \eta \end{matrix} \times \begin{matrix} x \\ y \end{matrix}$

$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \cdot \lambda_1 + \frac{\partial u}{\partial \eta} \cdot \lambda_2$ $\quad \frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}$

$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial \xi^2} \cdot \lambda_1^2 + 2\lambda_1\lambda_2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \lambda_2^2 \frac{\partial^2 u}{\partial \eta^2}$

$\bigcirc = \bigcirc$ $\quad \bigcirc = \bigcirc$

$$\Rightarrow (A\lambda_1^2+2B\lambda_1+C)\frac{\partial^2 u}{\partial\xi^2}+2[\lambda_1\lambda_2 A+(\lambda_1+\lambda_2)B+C]\frac{\partial^2 u}{\partial\xi\partial\eta}$$
$$+(A\lambda_2^2+2B\lambda_2+C)\frac{\partial^2 u}{\partial\eta^2}=0.$$

由于 $AC-B^2<0$ $A\neq 0$ ∴两不等实根

取 λ_1 λ_2 为两个实根

代入方程成 $\frac{\partial^2 u}{\partial\xi\partial\eta}=0$.

1000.13.75.

设 $z=z(u,v)$ 二阶偏，$z=z(x+y,x-y)$

满足：$\frac{\partial^2 z}{\partial x^2}+2\frac{\partial^2 z}{\partial x\partial y}+\frac{\partial^2 z}{\partial y^2}=1$

(1). 求 $z(u,v)$. $z<^{u=x+y}_{v=x-y}$

$$\begin{cases}\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot 1+\frac{\partial z}{\partial v}\cdot 1\\ \frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot 1-\frac{\partial z}{\partial v}\end{cases}$$

$$\frac{\partial^2 z}{\partial x^2}=\frac{\partial^2 z}{\partial u^2}+2\frac{\partial^2 z}{\partial u\partial v}+\frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial^2 z}{\partial y^2}=\frac{\partial^2 z}{\partial u^2}-2\frac{\partial^2 z}{\partial u\partial v}+\frac{\partial^2 z}{\partial v^2}$$

$$2\frac{\partial^2 z}{\partial x\partial y}=\cancel{2\frac{\partial^2 z}{\partial u\partial v}}+2\frac{\partial^2 z}{\partial u^2}-2\frac{\partial^2 z}{\partial v^2}$$

$$\Rightarrow 4\frac{\partial^2 z}{\partial u^2}=1$$

(2). 由(1)求出 $z=z(x+y,x-y)$ 一般表达式.

$$\frac{\partial^2 z}{\partial u^2}=\frac{1}{4}\qquad \frac{\partial z}{\partial u}=\frac{u}{4}+\varphi(v)$$

$$z=\frac{u^2}{8}+\int_0 \varphi(v)\,du+C_2(v)$$

$$=\frac{u^2}{8}+C_1(v)\cdot u+C_2(v)$$

$$=\frac{1}{8}(x+y)^2+C_1(x+y)(x-y)+C_2(x-y)$$